Foundation Pose Models for SLAM

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***Abstract*—Simultaneous Localization and Mapping (SLAM) has been fundamental in robotics and computer vision, enabling many advancements in robot navigation. Traditional SLAM methods have relied heavily on geometric constraints and feature-based approaches, which can struggle in texture-less and otherwise challenging scenes. The emergence of foundation models trained on large-scale vision datasets has opened new possibilities for SLAM and visual odometry. These models demonstrate robust and generalizable performance across diverse scenes with 0-shot performance. In this project we hope to take the first steps towards the integration of foundation pose models with SLAM systems. We hope to start by benchmarking state of the art vision pose foundation models against traditional SLAM baselines and ground truth motion capture data. Pending the adequate performance of an vision pose model we will incorporate this model into a larger visual inertia odometry system. We hope to achieve stable drone flight using only the foundation pose model and onboard drone sensing. Further potential improvements could include fine-tuning the foundation pose model on our specific data or focusing on achieving high accuracy enviromenmental mapping.**

***Index Terms*—SLAM, Bundle Adjustment, NeRF, Guassian Splat, ViT**

I. Introduction

Localization and mapping is a fundamental problem for robotic systems and is the baseline for almost all other navigation algorithms. The process by which a robot preforms this localization and mapping is known as SLAM. SLAM was first used successfully for vehicle localization and was introduced into the spotlight by its successful implementation is the DARPA urban challenge in 2000 [1]. Today's SLAM algorithms have improved significantly relying on complex feature matching with graph based optimizations. [2], [3]. Large foundation models have solved problem after problem in computer vision including classification [4], feature detection [5] and depth perception [6]. Over and over large feed forward networks have replaced traditional computer vision techniques offering high performance and faster execution. Recently camera pose estimation foundation models have emerged [7]–[9]. In this project we explore the feasibility of using this models and a replacement for traditional visual SLAM on the quadrotor system.

II. Related Work

A. Traditional SLAM

Simultaneous Localization and Mapping (SLAM) is a long standing problem in robotics [1]. Early SLAM methods relied on EKF like filtering, which evolved to more complex graph optimization based approaches. In MonoSLAM [3] real time monocular slam is accomplished by tracking features in a probabilistic framework. Methods like ORBSLAM [2] significancy increased performance on monocular and RGB-D scenes through robust feature matching and extraction algorithms. Methods like COLMAP [10], [11] rely on structure from motion (SfM) to extract camera intrinsics and extrinsics and intrisics as well as sparse point clouds. While COLMAP is not suitable for real time use, it produces highly accurate camera poses from only monocular information.

While modern SLAM methods are exceptionally powerfully, enabling everything from quadrotor flight to VR headsets, they have several limitations. Firstly they struggle in challenging scenes with changing lightening conditions or fast movements. Secondly their environmental reconstruction while geometrically accurate often lacks high visual accuracy which may be useful for downstream tasks.

B. Radiance Field SLAM

Radiance fields are an environment representation method which has recently gained prominence with the introduction of neural radiance fields (NeRFs) [12]. NeRFs allow a full environment to be reconstructed from a series of images. The resulting radiance field is highly visually accurate and can be used for many downstream tasks such as navigation [13] and manipulation [14]. The method of 3D Gaussian Splatting (3DGS) [15] has recently advanced the field by providing high-quality and high-speed training along with realtime rendering.

Due to the visually and geometrically expressive nature of these representations they are a naturally choice for SLAM. Many papers have combined NeRF with SLAM [16]–[19]. These works range from applying traditional SLAM pipelines soley to pose estimation [16] to hybrid approaches [18], [19], which leverage traditional SLAM optimization with the NeRF representation. 3DGS has more successfully been adapted to be used as a SLAM tool due to its much faster rendering and training [20], [21]. These approaches directly leverage camera extrinsic optimization through the differentiable visual representation itself. While radiance field based SLAM improves visual performance over traditional SLAM methods it is still quite slow. Even running on desktop grade hardware these radiance field base SLAM methods struggle to achieve real time due to the computational complexity of the radiance field

representation. However by utilizing vision foundation models trained on pose estimation we can achieve the expressiveness of NeRF with a single feed-forward pass.

C. Vision Foundation Models

Early vision foundation models were built on convolutional neural networks (CNNs), starting with the breakthrough ImageNet victory of AlexNet in 2012 [4]. This architecture was pushed further with the introduction of deep CNNs, often referred to as visual grounded geometry (VGG) networks, which implemented 16-19 layer stacks of 3 by 3 convolutions [22]. Traditionally these early image classification networks were supervised, meaning they required labeled training data, but self-supervised CNN frameworks such as SimCLR began to push the field towards unsupervised learning [23].

The field has since migrated toward using transformer backbones, the architecture first proposed by Google Deep Mind for translation but now known as the backbone for the boom of generative language models [24]. In 2021 the Google Brain team built upon this by introducing the first Vision Transformer (ViT) architecture, which replaced convolution with global self-attention over image patches. When pre-trained on sufficiently large image datasets, the ViT could match or surpass state-of-the-art CNNs on classification and transfer learning [25]. Subsequent self-supervised variants such as DINO scaled these ideas, with the DINOv2 network from Meta pushing the architecture to foundation model scale. The largest form of the network, using the ViT-g 1.1 billion parameter backbone, was trained without labels on a curated 142 million-image dataset, producing robust features that generalized across image distributions and tasks without the need for finetuning [5]. These transformer-based vision foundation models now provide the backbones for many contemporary 3-D perception pipelines including the ViT-based SLAM and multi-view geometry systems discussed in the next section. Their globally attentive, highly transferable features simplify camera pose, depth and scene understanding tasks that previously relied on handcrafted geometric modules.

Building on ViT backbones, recent work has begun to replace individual SLAM stages with neural-only modules: estimating the motion of up to 70k points over an extended period of time; monocular depth or full per-pixel geometry inferred in a single pass; and directly regressing NeRF-style 3D scenes from just one image [6], [26]–[28].

D. ViT SLAM

Today's state of the art has begun to treat SLAM itself as just another feed-forward vision problem. Instead of using the traditional pose graph optimization (PGO), or even chaining separate learned modules for tracking, mapping, and optimization, end-to-end ViT architectures now predict camera poses, per-view depth, and world-aligned point maps with dense tracking features in a single inference pass. Visual geometry grounded structure from motion (VGGSfM) introduced a fully differentiable SfM pipeline that could be trained end-to-end. The classical incremental pipeline of pair-wise matching, RANSAC camera registration, and bundle-adjustment is replaced with a point tracker module, a transformer camera-predictor that estimates all intrinsics and extrinsics, and a differentiable Levenberg-Marquardt layer to perform the bundle adjustment while still allowing gradients to flow through the network [7]. Pushing things further, dense and unconstrained stereo 3D reconstruction (DUSt3R) replaces the SfM chain with a pair-wise ViT that regresses a point-map for each image, which refers to a dense field of 3D points already expressed in the first view's coordinate frame. To extend from image pairs to scenes, DUSt3R drops the bundle adjustment in favor of a lightweight global alignment optimization that fits a single 3D similarity transform per image pair. Unlike VGGSfM which still need ground-truth camera intrinsics during training and relies on BA during inference, DUSt3R is trained purely on point-map regressions and infers both intrinsics and extrinsics from its own predictions [29].

The visual geometry grounded transformer (VGGT) makes the final abstraction to complete the fully feed-forward neural SLAM pipeline. Unlike DUSt3R/MASt3R, which reconcile pairwise predictions with global alignment, or VGGSfM, which still relies on differentiable bundle adjustment, VGGT removes virtually all geometry-specific post-optimization: a single forward pass simultaneously predicts calibrated cameras, per-view depth, world-aligned point maps and dense tracking features for up to hundreds of images, typically in $\leq$ 0.2s on an H100 GPU [8].

III. Proposed Work

We propose to integrate VGGT, a pose predicting ViT, with a odometry pipeline on a quadrotor. As a first step we will conduct real world experiments where we compare VGGT with baselines like DUSt3R, MASt3R and COLMAP along with ground truth motion capture data. This will allow us to determine if VGGT is accurate enough to be integrated into a SLAM pipeline. Furthermore this data will allow us to empirically determine the senor model which we will apply to the VGGT network, quantifying mean and covariance. We then hope to fuse the VGGT pose with existing inertia data available on the quadrotor to achieve stable flight.

We also consider two possible reach goals. First we could investigate fine-tuning VGGT on flight room specific data to improve performance. Since we have ground truth optitrack data we can fully supervise the pose output head. Furthermore we could possible utilize a much smaller network if we only need to achieve high performance within a single room. Second we can leverage the outputted point clouds to aid in both environmental reconstruction but also localization through iterative point cloud alignment. Overall we are perusing this as an exploratory work to evaluate the feasibility of a VGGT based SLAM pipeline. We consider failure to achieve the above goals to be a contribution in and of itself.

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